

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (CL) Examination

November-December 2012

CL204.01 Engineering Geology

Date: 30.11.2012, Friday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

## Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

## SECTION – I

- Q - 1 (a)** Define the following terms [07]  
(i) Oxbow lake (ii) Dune  
(iii) Epicenter and Focus
- Q - 2 (a)** Compare and contrast the following pairs with suitable examples: [06]  
(i) Lava and Magma (ii) Sills and Dykes  
(iii) Plutonic and Volcanic rocks
- (b)** Discuss briefly the mode of formation of Minerals [04]
- (c)** Describe the following rocks with reference to the civil engineering perspective. [04]  
(i) Basalt (ii) Shale  
(iii) Granite (iv) Marble

## OR

- Q - 2 (a)** “Structure and texture of Igneous rocks gives primary information regarding physio-chemical conditions of formation of the rock” Elaborate the statement. [08]
- (b)** What do you mean by *physical properties* of minerals? [02]
- (c)** What is *hardness* of mineral? Explain Moh’s Scale of Hardness. [04]
- Q - 3 (a)** Define the processes, *weathering, erosion, transportation* and *deposition*. [04]
- (b)** Describe different depositional features produced by the geological action of river. [06]
- (c)** Define rock cycle and explain the different processes involved in it. [04]

## OR

- Q - 3 (a)** What is earthquake? Describe different classifications and causes of earthquake. [08]
- (b)** Describe the interior structure of earth and major discontinuities? [06]

## SECTION – II

Q - 4 (a) What are the geological investigations carried out for selecting tunnel alignment? Add your note with tunnel in soft and hard rock formation. [07]

Q - 5 (a) With the help of neat diagram describe Arch dam and Gravity dam. [05]

(b) Write about the study of geological characters for dam site selection. Write a note on suitability of site with Anticline and Syncline type of fold for the dams. [09]

## OR

Q - 5 (a) Describe mass movement. Write note on external factors causing mass movement. [07]

(b) Define contour map, topographical map and geological map. How geological maps are helpful for civil engineering structures? [07]

Q - 6 (a) Draw the neat sketch for ideal fold and show its various parts. Define these parts. Differentiate Anticline and syncline fold. [07]

(b) With the help of neat diagram describe unconformities in geological strata? [07]

## OR

Q - 6 (a) Draw the neat sketch for ideal fault and show its various parts. Define these parts. Differentiate normal fault and reverse fault. [07]

(b) Define unconformities and joints in rocks. Add note on occurrence of joints and different types of joints. [07]

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